Chhaya

Problem Statement & Healthcare Use Case

HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026
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01 · CLINICAL CONTEXT & EPIDEMIOLOGICAL NEED

Problem Statement

About 101 million people in India live with diabetes (ICMR-INDIAB, 2023). Continuous glucose monitoring could guide their care, but one 14-day sensor costs roughly Rs 4,400 to 5,100. Wearing sensors all year would take about 26 of them, over Rs 1.1 lakh. So for many patients a sensor is worn once, a report is made, and the doctor treats from that report for months. Nobody tells the doctor how old the report has become, which of its readings not to believe, or what the patient's glucose probably did after the sensor came off.

DIABETIC BURDEN	14-DAY SENSOR COST	FULL ANNUAL WEAR	THE CLINICAL BLIND SPOT
101 Million	**₹4,400–5,100**	**> ₹1.1 Lakh**	**351 Days**
+136M prediabetes; 2nd highest in world (ICMR-INDIAB)	Single-use disposables; 100% out-of-pocket for families	26 sensors/yr; financially out of reach for >95% households	Over 96% of year unmonitored; doctors treat from stale reports

Given a health record and one sensor wear, Chhaya **(1)** estimates the patient's glucose after the sensor is removed, with an honest uncertainty band, **(2)** labels sensor lows that a fingerstick did not confirm, **(3)** prompts when the report has probably stopped describing the patient, and **(4)** proves every claim on patients the model never saw.

02 · CLINICAL DELIVERY & OPERATIONAL WORKFLOW

Healthcare Use Case

The setting is outpatient follow-up of Type 2 diabetes, where the doctor holds a one-time sensor report and occasional fingersticks. The user is the treating clinician. The decision supported is whether the existing sensor report can still be relied on, or whether to consider a new sensor wear.

This is "proactive, not reactive" scoped to what we measured. On the dashboard the doctor sees:

- **The patient's shadow:** an estimate with an 80% credible band drawn without the sensor, with the real sensor revealed on top;
- **Fingerstick-updated estimates:** running Bayesian innovation updates and the health record that sets the twin's prior;
- **The patient's own meal-excursion history:** empirical risk patterns during the calibration wear;
- **Scientific integrity ledger:** every result displayed directly beside the bar written before it was run.

`SAFE CLINICAL RESTRAINT: Chhaya gives no dose advice, no low-glucose estimate, and no automated alarm.`

03 · VERIFIED OUTCOMES ON HELD-OUT PATIENTS

What It Does Today

What it does today: Scored once on held-out patients against bars written beforehand, the estimate was **1.4 mg/dL closer** to the hidden sensor than the patient's own average day, five days after the sensor came off (19 CGMacros participants, 7 with type 2 diabetes, p = 4e-05). Fingersticks brought it **2.8 mg/dL closer** (29 Shanghai patients in supervised care, about six fingersticks a day). Of nine pre-registered bars, five passed and four missed, and all nine are published. The evidence comes from a Chinese and a US cohort; no Indian patient data was used. This is a research prototype, not for dosing.

04 · EXECUTIVE SUMMARY FOR PORTAL ENTRY

Short Version (If There's a Character Limit)

68 words · 495 characters

"101 million Indians live with diabetes, but a 14-day glucose sensor costs about Rs 4,400 to 5,100, so most patients wear one once and are treated from that report for months. Chhaya, a Type 2 diabetes digital twin, turns that one wear into a continuing estimate with an honest uncertainty band, labels unconfirmed sensor lows, and prompts the doctor when the report has probably gone stale. Every claim was scored once on held-out patients, and all nine pre-registered bars, passed and missed, are published."

Retail Price Verification Note: Sensor prices checked October 2026. Abbott FreeStyle Libre 1: MRP ₹4,671 / selling ₹4,437 on Truemeds. FreeStyle Libre 2: ₹4,624 to ₹5,138 in Amazon India price history. Used consistently across presentation deck, architecture, and video walkthrough.

Repository: github.com/AdityaChandel11/DigitalTwinHack
Live Demo: digital-twin-hack.vercel.app
Walkthrough Video: youtu.be/TOn2xY_z5L8